National Safety Code for Motor Carriers

Standard 10 – Cargo Securement

Version: June 2013

Standard 10

Cargo Securement

The NSC Standard 10 Cargo Securement was approved by the Council of Ministers Responsible for Transportation and Highway Safety on June 8, 2013.

The content and numbering of this text is identical to the Standard as approved by the Council of Ministers; however the appearance and layout may be different, including the pagination, the header and footer. This style has been adopted to be consistent with all standards that appear in the Manual: National Safety Code for Motor Carriers.

June 2013
ISBN 978-0-9878957-2-1

Table of Contents

INTRODUCTORY NOTES....10-6

INTERPRETATION....10-7

PART 1 - GENERAL PROVISIONS....10-11

Division 1 - Application....10-11
Application....10-11
Inspection....10-11
Division 2 - General Performance Criteria....10-13
Cargo securement systems....10-13
Performance criteria....10-13
Appropriate system....10-13
Equivalent Means of Securement....10-13
Division 3 - Requirements for Cargo Securement Systems....10-14
Exemption....10-14
General Requirement....10-14
Minimum Strength of Securement System....10-14
Securing Devices Marked With Working Load Limit....10-14
Unmarked Securing Devices....10-14
Steel strapping....10-15
Blocking Systems to Prevent Forward Movement....10-15
Rub rails....10-15
Material Used For Cargo Securement....10-15
Placement of Articles of Cargo....10-15
Prevention of Rolling....10-15
Division 4 - Tiedowns....10-16
Tension....10-16
Edge protectors....10-16
Working load limit....10-16
Minimum number....10-16
Division 5 - Front End Structures....10-18
Application....10-18
Height and width....10-18
Strength....10-18
Penetration resistance....10-18

PART 2 - SPECIFIC SECUREMENT REQUIREMENTS BY CARGO TYPE....10-19

Applicability....10-19
Division 1 - Logs....10-19
Application....10-19
Vehicle transporting logs....10-19
Log configuration....10-19
Securement System....10-20
Shortwood loaded crosswise....10-20
One stack of shortwood loaded crosswise....10-20
Two stacks of shortwood loaded crosswise....10-20
Long vehicles carrying shortwood loaded crosswise....10-21
Shortwood loaded lengthwise....10-21
Longwood loaded lengthwise....10-22

Pole trailers 10-22
Division 2 - Dressed Lumber 10-23
Application 10-23
Side by side 10-23
Securement system for dressed lumber 10-23
Bundles blocked against lateral movement by stakes 10-23
Bundles restrained from lateral movement by blocking or high friction devices 10-23
Bundles placed directly on top of other bundles or on spacers of adequate size and orientation 10-23
Layers of Bundles 10-24
Division 3 - Metal Coils 10-25
Application 10-25
Coils transported with eyes vertical by a vehicle or an intermodal container with anchor points 10-25
Coils transported with eyes crosswise by a vehicle or an intermodal container with anchor points 10-25
Coils transported with eyes lengthwise by a vehicle or an intermodal container with anchor points 10-26
Rows of Coils with Eyes Lengthwise 10-27
Prohibitions 10-28
Securement of Coils Transported in a Sided Vehicle or Intermodal Container without Anchor Points 10-28
Division 4 - Paper Rolls 10-29
Application 10-29
Friction mats 10-29
Chocks, wedges and blocking 10-29
Banding 10-29
Single layer of paper rolls transported eyes vertical in a sided vehicle 10-29
Split cargo of paper rolls transported with eyes vertical in a sided vehicle 10-30
Stacked cargo of paper rolls transported with eyes vertical in a sided vehicle 10-30
Single layer of paper rolls transported eyes crosswise in a sided vehicle 10-31
Rear doors 10-31
Stacked cargo of paper rolls transported eyes crosswise in a sided vehicle 10-31
Single layer of paper rolls transported eyes lengthwise in a sided vehicle 10-32
Stacked cargo of paper rolls transported eyes lengthwise in a sided vehicle 10-32
Cargo of Paper rolls transported with eyes vertical or with eyes lengthwise on a flatbed vehicle or in a curtain sided vehicle 10-33
Cargo of Paper rolls transported with eyes crosswise on a flatbed vehicle or in a curtain sided vehicle 10-33
Division 5 -Concrete Pipe 10-34
Application 10-34
Tiedowns 10-34
Blocking 10-34
Multi-sized pipe 10-34
Arranging layers 10-34
Bell Pipe 10-35
Single layer or Bottom layer, small pipe 10-35
Tiedowns for securing layers of small concrete pipes 10-36
Top layer, small pipes 10-36
Large pipes 10-36
Division 6 - Intermodal Containers 10-38
Application 10-38
Intermodal container transported on Container chassis vehicle 10-38

Loaded intermodal container transported on other vehicles 10-38
Empty intermodal container transported on other vehicles 10-38
Division 7 - Vehicles as Cargo 10-40
Application 10-40
Light vehicles 10-40
Heavy vehicles 10-40
Flattened or crushed light vehicles 10-40
Prohibition 10-41
Securement system for immobilizing stacks of flattened or crushed vehicles 10-41
Containment of Loose Parts 10-41
Division 8 - Roll-on/roll-off and Hook Lift Containers 10-42
Application 10-42
Replacement of Securing Devices 10-42
No integral securement system 10-42
Division 9 - Boulders 10-43
Application 10-43
Positioning on vehicle 10-43
Tiedowns 10-43
Number of tiedowns 10-44

PART 3 - Default Working Load Limits 10-45

PART 4 - Manufacturing Standards 10-46

INTRODUCTORY NOTES

The standard was drafted with the objective to provide jurisdictions with a standard which can be adopted by reference.

Each jurisdiction may encounter situations where existing legislation or regulations contain definitions or provisions which conflict or overlap with the contents of this Standard. For this reason, the following sections are offered as optional text for consideration by jurisdictions where needed:

Interpretation (Definitions)

"driver" includes the owner of the vehicle if the owner of the vehicle is not the carrier

PART 1 - GENERAL PROVISIONS

Division 1 - Application

Safety responsibility

(1) A carrier shall not permit a driver to operate a vehicle where the cargo transported in or on the vehicle is not contained, immobilized or secured in accordance with this Standard.

(2) A driver shall not operate a vehicle where the cargo transported in or on the vehicle is not contained, immobilized or secured in accordance with this Standard.

Division 2 - General Performance Criteria

Cargo securement

Cargo transported by a vehicle shall be contained, immobilized or secured so that it cannot

(a) leak, spill, blow off, fall from, fall through or otherwise be dislodged from the vehicle, or

(b) shift upon or within the vehicle to such an extent that the vehicle's stability or manoeuvrability is adversely affected

INTERPRETATION

In this Standard,

"anchor point" means the part of the structure, fitting or attachment on a vehicle or cargo to which a tiedown is attached;

"bell pipe" means concrete pipe with a flanged end that is bigger in diameter than the barrel;

"blocking" means a substantial structure, device or article placed against or around cargo to prevent horizontal movement;

"bolster" means a transverse, load bearing, structural horizontal component of a bunk securing device;

"boulder" means a single piece of natural or quarried, irregularly shaped rock

(a) that weighs 5 000 kilograms or more, or

(b) that has a volume of more than 2 cubic metres;

"bracing" means a structure, device or article placed against another structure, device or article to prevent tipping;

"bulkhead" means a vertical barrier across a vehicle to prevent the cargo moving forward;

"bundle" means articles that have been unitized for the purpose of securing them as a single article with a uniform shape;

"bunk" means a horizontal bolster that

(i) is installed transversely across a vehicle, and

(ii) is fitted with a stake at each end;

"cab shield" means a vertical barrier

(i) placed directly behind the cab of a truck or truck tractor, and

(ii) capable of protecting the driver if cargo moves forward;

"cargo" means all articles or material carried by a vehicle, including those used in the operation of the vehicle

"cargo securement system" means the method by which cargo is contained or secured and includes vehicle structures, securing devices and all components of the system

"chock" means a tapered or wedge-shaped part used to prevent round articles from rolling

"cleat" means a short piece of material nailed to the deck to reinforce blocking;

"coil bunk" means a device that keeps the timbers supporting a metal coil in place;

"contained" with respect to cargo means that

(i) the cargo fills a sided vehicle,

(ii) every article is in contact with or close to a wall or other articles, and

(iii) the cargo cannot move or tip;

“container chassis vehicle” means a vehicle specifically built for and fitted with locking devices for the transport of intermodal containers;

“cradle” means a structure that holds a circular article and prevents it from rolling;

“cylinder well” means the depression formed between 2 cylindrical articles when they are laid against each other with their eyes horizontal and parallel;

“deck” means the floor of a vehicle onto which the cargo is loaded;

“dunnage” means loose material used to support and protect cargo;

"duty status" means, in respect of a driver, any of the following periods:

(i) off-duty time spent in a sleeper berth;

(ii) off-duty time, other than time spent in a sleeper berth;

(iii) driving time; or

(iv) on-duty time, other than driving time.

“edge protector” means a device put on the exposed edge of an article of cargo

(i) to protect a tiedown or the article from damage, or

(ii) to distribute tiedown forces over a greater area;

“flatbed vehicle” means a vehicle with a deck but no permanent sides;

“forward” with respect to a vehicle means towards the cab or engine;

“frame vehicle” means a vehicle for transporting logs that has a skeletal structure fitted with a front bunk and a rear bunk that together cradle a stack of logs as an integral part of the vehicle;

“friction mat” means a device placed between a deck and cargo, or between articles of cargo, that increases the friction between them;

"front end structure" means a vertical barrier across the front of a deck that prevents cargo moving forward;

“heavy vehicle” means

(i) a vehicle that weighs more than 4 500 kilograms, or

(ii) equipment or machinery that operates on wheels or tracks and weighs more than 4 500 kilograms;

“Hook-lift Container” means a specialized container that is loaded and unloaded onto a tilt frame body by an articulating hook-arm.

“integral locking device “ means a device that is designed and used to restrain an article of cargo by

connecting and locking attachment points on the article to anchor points on the vehicle;

"integral securement system" means a roll-on/roll-off container or a Hook-lift Container and the vehicle used to transport them which are equipped with compatible front and rear hold-down devices which secure the container to the vehicle;

"intermodal container" means a reusable, transportable container that is specially designed with integral locking devices to secure it to a container chassis vehicle;

"large pipe" means concrete pipe with an inside diameter of more than 114.3 centimetres;

"lengthwise" means along the vehicle from the cab or tongue at the front to the rear;

"light vehicle" means an automobile, truck or van that weighs 4 500 kilograms or less;

"log" includes a utility pole, a treated pole and a building component of a log cabin;

"longwood" means logs longer than 4.9 m;

"pallet" means a platform or tray on which cargo is placed so that it can be handled as a unit;

"pole trailer" means a trailer with a frame that consists only of a drawbar;

"rail vehicle" means a vehicle fitted with stakes at the front and rear to contain logs loaded crosswise;

"restrain" includes prevent from tipping or moving;

"Roll-on / Roll-off Container" means a specialized container which is loaded and unloaded onto a tilt frame body by a lifting mechanism in conjunction with rollers which are fixed to the container.

"rub rail" means a rail along the side of a vehicle that protects the side of the vehicle from impact;

"securing device" means a device specifically designed and manufactured to attach, restrain or secure cargo;

"shoring bar " means a device placed transversely between the walls of a vehicle and cargo to prevent the cargo from tipping or moving;

"shortwood" means logs that are not more than 4.9 metres long;

"sided vehicle" means a vehicle, including a van, a dump bodied vehicle and a sided intermodal container carried by vehicle, with a cargo compartment that is enclosed on all sides by walls

(i) that are strong enough to contain the cargo, and

(ii) that may have latched openings for loading and unloading;

"small pipe" means concrete pipe with an inside diameter of up to 114.3 centimetres;

"spacer" means material placed under an article, or between layers of articles, to make loading and unloading easier;

"stake" means a part, including a standard, that

(i) is mounted close to vertical on a vehicle frame or as part of a bunk, and

(ii) that immobilizes cargo placed against it;

“strapping” means tensioned strips of material that are clamped or crimped back on themselves;

“tiedown” means a combination of securing devices that are attached to one or more anchor points on a vehicle;

"transport" means the carriage of cargo by a vehicle while on a highway

“unitized” means wrapped, banded or bound together so that several articles can be handled as a single article of cargo or behave as a single article;

“vehicle” means a truck, a truck tractor, individually or in combination with one or more semitrailers or trailers

“void filler” means material that is

(i) used to fill a space between the cargo and the structure of the vehicle, and

(ii) is strong enough to prevent the cargo from moving;

“working load limit” means the maximum load that may be applied to a component of a cargo securement system during normal service determined in accordance with Divisions 3 and 4 of Part 1.

PART 1 - GENERAL PROVISIONS

Division 1 - Application

Application

2(1) This Standard applies to a vehicle or combination of vehicles

(a) transporting cargo on a highway, and

(b) exceeding a registered gross vehicle weight of 4 500 kilograms.

(2) This Standard applies when an intermodal container is used to transport cargo.

Inspection

3(1) The driver of a vehicle shall

(a) inspect the vehicle to confirm that the vehicle's tailgate, tailboard, doors, tarpaulins and spare tire, and other equipment used in its operation, are secured,

(b) ensure that the cargo does not interfere with the driver's ability to drive the vehicle safely, and

(c) ensure that the cargo does not interfere with the free exit of a person from the cab or driver's compartment of the vehicle.

(2) The driver of a vehicle shall inspect the vehicle's cargo and the cargo securement system used and make necessary adjustments:

(a) before driving the vehicle, and

(b) not more than 80 kilometres from the point where the cargo was loaded,

(3) The driver of a vehicle shall regularly re-inspect the vehicle's cargo and the cargo securement systems used and make adjustments to the cargo or cargo securement system as necessary, including adding more securing devices, at intervals based on whichever of the following occurs first:

(a) there is a change of duty status of the driver,

(b) the vehicle has been driven for 3 hours; or

(c) the vehicle has been driven for 240 kilometres.

(4) Subsections (2) and (3) do not apply to a driver where

(a) the cargo is sealed in a vehicle and the driver has been ordered not to open it to inspect the cargo, or

(b) the vehicle is loaded in a manner that makes the cargo, or portions of the cargo, inaccessible.

(5) The driver of a vehicle transporting logs, before the vehicle enters a highway from a private road, shall:

(a) inspect the vehicle, the logs and the securing devices to ensure compliance with this Standard, and

(b) make necessary adjustments to the securing devices, including adding more securing devices.

Division 2 - General Performance Criteria

Cargo securement systems

4(1) Cargo securement systems, and each component of a system, used to contain, immobilize or secure cargo on or within the vehicle shall be strong enough to withstand the forces described in section 5(1).

(2) The components of the cargo securement system of a vehicle.

(a) shall be in proper working order,

(b) shall be fit for the purpose for which they are used,

(c) shall have no knots, damaged or weakened components that will adversely affect their performance for cargo securement purposes, and

(d) shall not have any cracks or cuts.

(3) A securing device, integral locking device, movable structure or blocking device used to secure cargo to a vehicle shall itself be secured in a manner that prevents it from becoming unfastened while the vehicle is on a highway.

Performance criteria

5(1) The cargo securement system shall be capable of withstanding the forces that result if the vehicle is subjected to each of the following accelerations:

(a) 0.8 g deceleration in a forward direction;

(b) 0.5 g deceleration in a rearward direction;

(c) 0.5 g acceleration in either sideways direction.

(2) The cargo securement system shall provide a downward force equal to at least 20 % of the weight of an article of cargo if the article is not fully contained within the structure of the vehicle.

(3) The load on a component of a cargo securement system that reacts to a force referred to in subsection (1) or (2), shall not exceed the working load limit of the component.

Appropriate system

6(1) The cargo securement system used to contain, immobilize or restrain cargo shall be appropriate for the size, shape, strength and characteristics of the cargo.

(2) The securing devices used to secure cargo on or within a vehicle shall be

(a) designed and constructed for the purpose for which they are used, and

(b) used and maintained in accordance with the manufacturer's instructions.

Equivalent Means of Securement

7 Where cargo transported by a vehicle is contained, immobilized or secured in accordance with the applicable requirements of Divisions 3, 4 and 5 and Part 2, it meets the requirements of Section 5.

Division 3 - Requirements for Cargo Securement Systems

Exemption

8 This Division does not apply to the transportation of a commodity that is transported in bulk, lacks structure, fixed shape or is fluid, and is transported in

(a) a tank, hopper, box or container that is manufactured as part of the vehicle in which it is transported, or

(b) a vehicle that is specifically manufactured to transport the commodity.

General Requirement

9. Cargo shall be firmly immobilized or secured on or within a vehicle by structures of adequate strength, blocking, bracing, dunnage or dunnage bags, shoring bars, tiedowns or a combination of these.

Minimum Strength of Securement System

10 (1) In this section, the "aggregate working load limit" is the sum of One-half of the working load limit for each end section of a tiedown that is attached to an anchor point.

(2) The aggregate working load limit of the cargo securement system used to secure an article of cargo on or within a vehicle shall not be less than 50% of the weight of the article.

(3) The aggregate working load limit of the cargo securement system used to secure a group of articles of cargo on or within a vehicle shall be not less than 50% of the total weight of the group.

Securing Devices Marked With Working Load Limit

11(1) The working load limit of a tiedown or a component of a tiedown that is marked by its manufacturer with a numeric working load limit is the marked working load limit.

(2) A tiedown or a component of a tiedown that is marked by its manufacturer in accordance with a standard referred to in Part 4 has a working load limit equal to that standard.

(3) A chain that is marked by the manufacturer in accordance with the table of Working Load Limits under Part 4 – Section 7 has a working load limit equal to the amount shown for the grade and size.

(4) A person shall not use a tiedown or a component of a tiedown to secure cargo to a vehicle unless it is marked by the manufacturer with respect to its working load limit.

Unmarked Securing Devices

12(1) *Deleted September 2010*

(2) *Deleted September 2010*

(3) *Deleted September 2010.*

(4) *Deleted September 2010.*

(5) *Deleted September 2010.*

(6) *Deleted September 2010.*

(7) *Deleted September 2010.*

(8) *Deleted September 2010.*

(9) To be considered part of a cargo securement system, a friction mat must be marked by its manufacturer with the maximum usable friction resistance (in g's) the mat will provide in restraining cargo against horizontal and lateral movement.

(10) *Deleted September 2010.*

Steel strapping

13(1) Steel strapping that is 2.54 centimetres wide or wider used to secure cargo to a vehicle shall have at least 2 pairs of crimps in each seal

(2) An end-over-end lap joint formed in steel strapping used to secure cargo to a vehicle shall be sealed with at least 2 seals.

Blocking Systems to Prevent Forward Movement

14 The aggregate working load limit of the components of a blocking system used as a unique form of securement to prevent an article of cargo from moving forward, including tiedowns used as blocking, shall not be less than 50% of the weight of the article being blocked.

Rub rails

15 The securing devices used on or within a vehicle shall, wherever practical, be located inboard the rub rails where the vehicle has rub rails.

Material Used For Cargo Securement

16 Material used on or within a vehicle as dunnage, chocks or cradles or for blocking or bracing shall be strong enough that it will not be split or crushed by the cargo or the tiedowns.

Placement of Articles of Cargo

17 Where the articles of cargo on or within a vehicle are placed beside each other and secured by tiedowns that pass over 2 or more articles, the articles shall be

(a) placed in direct contact with each other, or

(b) prevented from moving towards each other while the vehicle is on a highway.

Prevention of Rolling

18 Where any cargo or portion thereof may roll, it shall be restrained by chocks, wedges, a cradle or another securing device that prevents the cargo from rolling.

Division 4 - Tiedowns

Tension

19(1) A tiedown used to secure cargo to a vehicle shall be designed, constructed and maintained so that the driver of the vehicle can tighten it.

(2) The driver of a vehicle shall ensure that tiedowns are taut while the vehicle is on a highway.

(3) Subsection (1) does not apply to steel strapping.

(4) A tiedown used to secure cargo to a vehicle shall be used in a manner that prevents the tiedown from slipping, loosening, unfastening, opening or releasing while the vehicle is on a highway.

(5) A tiedown used to secure stacked articles of cargo is considered to contribute to the securement of all articles of cargo on which that tiedown causes pressure.

Edge protectors

20(1) An edge protector shall be used where a tiedown would be subject to abrasion or cutting at the point where it touches an article of cargo.

(2) An edge protector used between a tiedown and cargo shall be resistant to abrasion, cuts and crushing.

(3) An edge protector used between a tiedown and cargo shall allow the tiedown to slide freely when it is tightened or loosened.

Working load limit

21(1) Subject to subsection (2), the working load limit of a tiedown, associated connector or attachment mechanism is the lesser of

(a) the lowest working load limit of the components, or

(b) the lowest working load limit of the anchor points or associated connector or attachment mechanism to which the tiedown is attached.

(2) The working load limit of a tiedown that includes synthetic webbing is the least of

(a) the working load limit of the synthetic webbing assembly, or

(b) the lowest working load limit of the components, or

(c) the working load limit of the anchor point to which the tiedown is attached.

Minimum number

22(1) Subject to subsection (4), cargo transported by a vehicle shall be secured using the number of tiedowns calculated under subsection (2) or (3).

(2) Where an article of cargo is not blocked or immobilized by a front end structure, bulkhead, by other immobilized cargo or by another device that prevents it moving forward, it shall be secured by at least

(a) 1 tiedown where the article is 1.52 metres or shorter and weighs not more than 500 kilograms,

(b) 2 tiedowns where the article is

(i) 1.52 metres or shorter and weighs more than 500 kilograms, or

(ii) longer than 1.52 metres but not longer than 3.04 metres regardless of its weight, or

(c) where the article is longer than 3.04 metres

(i) 2 tiedowns for the first 3.04 metres of length, and

(ii) 1 extra tiedown for each additional 3.04 metres or fraction of 3.04 metres.

(3) Where an article of cargo is blocked or immobilized by a front end structure, bulkhead, by other immobilized cargo or by another device to prevent it moving forward, it shall be secured by at least

(a) 1 tiedown where the article is not longer than 3.04 meters, or

(b) where the article is longer than 3.04 metres

(i) 1 tiedown for the first 3.04 metres of length, and

(ii) 1 extra tiedown for each additional 3.04 metres or fraction of 3.04 metres.

(4) Where a vehicle is transporting machinery or fabricated structural items that shall be secured by special methods because of their design, size, shape or weight, the special methods shall:

(a) secured adequately any article of the cargo

(b) be properly used in accordance with the manufacturer's instructions.

Division 5 - Front End Structures

Application

23(1) Subject to subsection (2), this Division applies to a vehicle transporting cargo that is in contact with the front-end structure of the vehicle.

(2) Where devices are used on a vehicle that perform the same function as the front end structure of a vehicle, the devices shall be at least as strong as and provide the same protection as a front end structure that complies with this Division.

(3) A cab shield is not a front end structure or part of a cargo securement system.

Height and width

24(1) The height of the front end structure of the vehicle shall not be shorter than the shorter of

(a) the height at which it prevents the cargo from moving forward, and

(b) 122 centimetres above the deck.

(2) The width of the front end structure of a vehicle shall not be narrower than the narrower of

(a) the width of the vehicle, and

(b) the width at which it prevents the cargo being transported from moving forward.

Strength

25(1) The front end structure of a vehicle shall be able to withstand a horizontal forward static load equal to 50% of the total weight of the cargo where

(a) the height of the front end structure is shorter than 1.83 metres, and

(b) the cargo is uniformly distributed over all of the front end structure.

(2) The front end structure of a vehicle shall be able to withstand a horizontal forward static load equal to 40% of the total weight of the cargo where

(a) the height of the front end structure is 1.83 metres or higher, and

(b) the cargo is uniformly distributed over all of the front end structure.

Penetration resistance

26(1) The front-end structure of the vehicle shall be able to resist penetration by an article of cargo that contacts it when the vehicle decelerates at a rate of 6.1 metres per second per second.

(2) The front-end structure of the vehicle shall not have an opening or gap that is big enough to permit an article of cargo to pass through it.

PART 2 - SPECIFIC SECUREMENT REQUIREMENTS BY CARGO TYPE

Applicability

27(1) This Part applies in addition to and not instead of Part 1.

(2) Where a requirement for containing, immobilizing or securing cargo transported by a vehicle required under this Part differs from a requirement under Part 1, the provisions of this Part apply.

Division 1 - Logs

Application

28(1) This Division applies to the transportation of logs that

(a) *Deleted September 2010*

(b) are part of a cargo that has more than 4 processed logs.

(2) This Division does not apply to firewood, stumps, log debris or logs that are transported in a vehicle or container that is enclosed on all sides and strong enough to contain them.

(3) A cargo of 4 or less processed logs loaded lengthwise may be secured by the provisions of this Division or the general securement provisions of Part 1.

Vehicle transporting logs

29(1) A vehicle that is transporting logs shall be designed, built or specially adapted for such transportation.

(2) The vehicle shall be fitted with bunks, bolsters, stakes or other means of cradling the logs and preventing them from shifting.

(3) Stakes that are not permanently attached to the vehicle frame or bunk shall be secured in a manner that prevents the stakes from separating from the vehicle while it is on a highway.

Log configuration

30(1) Logs shall be solidly packed on a vehicle.

(2) The outer logs in the bottom layer of logs shall touch and rest solidly against a bunk, bolster or stake.

(3) Outside logs on a stack of logs shall

(a) touch at least 2 bunks, bolsters or stakes, or

(b) where one end of a log does not touch a bunk, bolster or stake, it shall

(i) rest on other logs in a stable manner, and

(ii) extend beyond the end of the bunk, bolster or stake.

(4) The centre of the highest outside log on each side or end of the vehicle shall be lower than the tops of the bunks or stakes.

(5) The upper logs that form the top of the cargo shall be crowned.

Securement System

31 Tiedowns must be used to secure the load, in combination with bunks, bolsters, stakes, or other means of cradling the logs.

32 Sections 10(2) and 10(3) do not apply to this Division.

33 Sufficient additional tiedowns or other securing devices shall be used to ensure that no part of the cargo becomes dislodged where

(a) the wood's condition results in such low friction between logs that they may slip against each other, or

(b) a log is not held in place by contact with other logs or by the bunks, bolsters or stakes.

Shortwood loaded crosswise

34(1) This section and sections 35 to 37 apply to shortwood loaded crosswise on a frame, rail or flatbed vehicle other than a pole trailer.

(2) The end of a log in the lower layer of shortwood shall not extend more than 1/3 of the log's total length beyond the nearest supporting structure on the vehicle.

One stack of shortwood loaded crosswise

35(1) Despite section 22, where only one stack of shortwood is loaded crosswise, the stack shall be secured by at least 2 tiedowns arranged as follows:

(a) The tiedowns shall attach to the vehicle frame at the front and rear of the load, and shall cross the load lengthwise;

(b) The tiedowns shall be positioned at approximately one-third and two-thirds of the length of the logs;

(2) A vehicle built on or after January 1, 2010 shall be equipped with a device that maintains a tension not less than 900 kg at all times, and automatically takes up slack in the tiedown as the logs settle.

Two stacks of shortwood loaded crosswise

36(1) Despite section 22, where two stacks of shortwood are loaded crosswise and side-by-side on a vehicle, they shall be loaded so that

(a) there is no space between the 2 stacks,

(b) the outside of each stack is raised by a piece of metal at least 25 millimetres high within 10 centimetres of the end of the logs or the side of the vehicle and causes the load to lean toward the centre to the vehicle,

(c) the highest log is not more than 2.44 metres above the deck, and

(d) at least one tiedown used lengthwise across each stack shall

(i) be located approximately midway between the bunks or stakes, and

(ii) attach to the vehicle frame at the front and rear of the load.

(2) A vehicle built on or after January 1, 2010 shall be equipped with a device that maintains a tension not less than 900 kg at all times, and automatically takes up slack in the tiedown as the logs settle.

Long vehicles carrying shortwood loaded crosswise

37(1) A vehicle with a cargo carrying surface that is more than 10 metres long transporting shortwood loaded crosswise shall have centre stakes, or comparable structures, that divide its length into two approximately equal sections.

(2) Where the vehicle is divided by centre stakes, each tiedown shall

(a) secure the highest log on each side of the centre stake, and

(b) be fastened below that highest log.

(3) Where the vehicle is divided by centre stakes, each tiedown shall

(a) be fixed at each end and tensioned from the middle,

(b) be fixed in the middle and tensioned from each end, or

(c) pass through a pulley or similar device in the middle of the tiedown and be tensioned from one end.

(4) Where a stake or other structure on a vehicle transporting shortwood loaded crosswise is subjected to an upward force when the tiedowns are tightened, the stake or other structure shall be anchored to resist that force.

Shortwood loaded lengthwise

38(1) Despite section 22, each stack of shortwood loaded lengthwise on a frame vehicle or flatbed vehicle, other than a pole trailer, shall be secured to the vehicle by 2 or more tiedowns.

(2) Despite subsection (1), a stack of shortwood loaded lengthwise on a frame vehicle or flatbed vehicle, other than a pole trailer, shall be secured to the vehicle with a single tiedown located approximately midway between the bunks or stakes where all the logs in the stack

(a) are shorter than 3.04 metres,

(b) are blocked in the front by a front end structure strong enough to restrain the cargo or by another stack, and

(c) are blocked in the rear by another stack or the vehicle's end structure.

(4) The aggregate working limit of tiedowns used to secure each stack shall be at least 1/6 of the weight of the stack

Longwood loaded lengthwise

39(1) Despite section 22, a stack of longwood loaded lengthwise on a frame or flatbed vehicle, other than a pole trailer, shall be secured to the vehicle by 2 or more tiedowns.

(2) The aggregate working limit of tiedowns used to secure each stack shall be at least 1/6 of the weight of the stack

(3) The outside logs of a stack of longwood shall be secured by 2 or more tiedowns.

Pole trailers

40(1) This section applies to logs, regardless of the length of individual logs, transported on pole trailers.

(2) Despite section 22, the logs shall be secured by

(a) one or more tiedowns at each bunk, or

(b) two or more tiedowns used as wrappers that encircle the entire stack of logs at sufficient locations along the stack to secure it effectively.

(3) Where wrappers are used on a stack of logs, the wrappers at the front and rear ends of the stack shall be not less than 3.04 metres apart.

(4) Where the vehicle is transporting one or two logs with diameters greater than 0.6 metre, each log shall be individually immobilized with chock blocks or an equally effective method that prevents the logs from moving.

(5) Where a log with a diameter greater than 0.6 metre rises above the bunks, it shall be secured to the underlying logs with at least 2 additional tiedowns used as wrappers.

Division 2 - Dressed Lumber

Application

41(1) This Division applies to the transportation of

(a) bundles of dressed lumber and packaged lumber, and

(b) unitized building products, including plywood, gypsum board or other materials of similar shape.

(2) For the purpose of this Division, "bundle" means the material referred to in subsection (1), but does not include building products loaded on pallets or packages of engineered wood products such as beams or trusses.

(3) Dressed lumber and similar bundled building products being transported in enclosed trucks or trailers can be secured

(a) in accordance with the requirements of this Division, or

(b) the general securement provisions of Part 1.

Side by side

42 Where bundles are placed side by side

(a) bundles shall be in direct contact with each other, or

(b) a method shall be used that prevents the bundles from moving towards each other.

Securement system for dressed lumber

43. Bundles carried in 2 or more layers shall be secured in accordance with one of sections 44, 45, 46 or 47

Bundles blocked against lateral movement by stakes

44 Bundles carried in 2 or more layers which are blocked against lateral movement by stakes on the sides of the vehicle shall be secured by tiedowns laid out over the top layer, as outlined in the provisions of section 22 of this Standard

Bundles restrained from lateral movement by blocking or high friction devices

45 Bundles carried in 2 or more layers which are restrained from lateral movement by blocking or high friction devices between layers shall be secured by tiedowns laid out over the top tier, as outlined in the provisions of section 22 of this Standard

Bundles placed directly on top of other bundles or on spacers of adequate size and orientation

46(1) Bundles carried in 2 or more layers placed directly on top of other bundles or on spacers of adequate size and orientation, shall be secured by.

(a) tiedowns over the top layer of bundles, in accordance with provisions of section 22 of this Standard, with a minimum of two tiedowns for bundle(s) longer than 1.52 metres, and

(b) tiedowns over the middle layer of bundles in accordance with the provisions of section 22 of this Standard, for each stack of bundles higher than 1.85 metres composed of more than two layers; and

(2) where spacers are used, then

(a) the length of spacers between bundles shall provide support to all pieces in the bottom row of the bundle, and.

(b) the width of individual spacers shall be equal to or greater than the height, and

(c) spacers shall provide good interlayer friction, and

(d) where spacers are comprised of layers of material, the layers shall be unitized or fastened together in a manner which ensures that the spacer performs as a single piece of material.

Layers of Bundles

47. Bundles carried in 2 or more layers shall be secured by tiedowns laid out over each layer of bundles, in accordance with the provisions of section 22 of this Standard with a minimum of two tiedowns over each top bundle(s) longer than 1.52 metres, in all other circumstances.